

Continuous Kantorovich Formulation

Exercise statements

This version contains the exercises only. Hints and worked solutions are available in the companion PDF.

Exercise 1 (Couplings in basic cases). (a) Let α, β be finite *positive* measures with *different* total masses. What is $\Pi(\alpha, \beta)$, the set of couplings?

- (b) (i) What are the couplings between δ_x and $\mathcal{U}[0, 1]$?
(ii) What are the couplings between $\frac{1}{2}(\delta_x + \delta_y)$ and $\mathcal{U}[0, 1]$, assuming $x \neq y$?
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Exercise 2 (Gaussian couplings in $\mathbb{R}$ (zero means)).

Let $\alpha = \mathcal{N}(0, \sigma_\alpha^2)$ and $\beta = \mathcal{N}(0, \sigma_\beta^2)$ with $\sigma_\alpha, \sigma_\beta \geq 0$. Consider *Gaussian* couplings $\pi = \mathcal{N}(0, \Sigma)$ on $\mathbb{R} \times \mathbb{R}$ with

$$\Sigma = \begin{pmatrix} \sigma_\alpha^2 & c \\ c & \sigma_\beta^2 \end{pmatrix} \geq 0.$$

- (a) Express the quadratic OT cost $\mathbb{E}[(X - Y)^2]$ in terms of c .
(b) Determine the constraint on c implied by $\Sigma \geq 0$.
(c) Solve the Kantorovich problem *restricted to Gaussian couplings* by minimizing over feasible c .
(d) What is the rank of Σ as a function of $(\sigma_\alpha, \sigma_\beta, c)$, and what is the support of π in each case? Compare the optimal Gaussian coupling with the conclusion of Brenier's theorem.
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Exercise 3 (Uniform $\rightarrow$ two Diracs; two Diracs $\rightarrow$ uniform).

Fix two distinct points $x \neq y$ in $\mathbb{R}$ and let

$$\alpha = \mathcal{U}[0, 1], \quad \beta = \frac{1}{2} \delta_x + \frac{1}{2} \delta_y.$$

Consider the quadratic cost $c(u, v) = |u - v|^2$ (the increasing rearrangement remains optimal for $|u - v|^p$, $p \geq 1$, but uniqueness may fail for $p = 1$).

- (a) Find the $\mathcal{W}_p$ -optimal Monge map $T : [0, 1] \rightarrow \mathbb{R}$ that pushes α to β .
(b) Is there a Monge map $S : \mathbb{R} \rightarrow [0, 1]$ such that $S_\# \beta = \alpha$?
(c) Describe an optimal Kantorovich coupling $\pi \in \Pi(\alpha, \beta)$ between α and β .
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Exercise 4 (Bottleneck cost: metric property and closed-form OT value).

Let $(X, \mathcal{B})$ be a standard Borel space (so the diagonal in $X \times X$ is measurable) and let $m : X \rightarrow (0, \infty)$ be measurable (strictly positive everywhere). Define

$$c(x, y) := \begin{cases} 0, & x = y, \\ m(x) + m(y), & x \neq y. \end{cases}$$

- (a) Show that $d(x, y) := c(x, y)$ is a metric on X .
(b) For probability measures α, β on $(X, \mathcal{B})$ with $\int m d\alpha + \int m d\beta < \infty$, compute the Kantorovich value

$$W_c(\alpha, \beta) := \inf_{\pi \in \Pi(\alpha, \beta)} \int c(x, y) d\pi(x, y)$$

in closed form.

- (c) Give an explicit optimal plan in terms of the maximal common submeasure $\mu := \alpha \wedge \beta$.

Further exercises

Exercise 5 (Gaussian couplings in $\mathbb{R}^d$ (zero means)).

Let $\alpha = \mathcal{N}(0, \Sigma_\alpha)$ and $\beta = \mathcal{N}(0, \Sigma_\beta)$ with $\Sigma_\alpha, \Sigma_\beta \in \mathbb{S}_{++}^d$. Any Gaussian coupling on $\mathbb{R}^d \times \mathbb{R}^d$ has block covariance

$$\Sigma = \begin{pmatrix} \Sigma_\alpha & K \\ K^\top & \Sigma_\beta \end{pmatrix} \geq 0.$$

- (a) Express the quadratic OT cost $\mathbb{E}\|X - Y\|^2$ in terms of K .
- (b) Determine the constraint on K implied by $\Sigma \geq 0$ (give equivalent forms).
- (c) Solve the Kantorovich problem *restricted to Gaussian couplings* by minimizing the cost over feasible K .
- (d) What is the rank of Σ and the support of the coupling π for (i) strict feasibility and (ii) an optimizer from part (c)? Compare the optimal Gaussian coupling with Brenier's theorem.

Exercise 6 (A non-Monge optimal plan in $\mathbb{R}^2$).

Let α be the uniform probability measure on the vertical segment

$$S_0 := \{(0, s) : s \in [0, 1]\}$$

(with respect to 1D length, normalized), and let

$$\beta := \frac{1}{2} \mathcal{U}(S_-) + \frac{1}{2} \mathcal{U}(S_+), \quad S_\pm := \{(\pm 1, t) : t \in [0, 1]\}.$$

We consider the quadratic cost $c(x, y) = \|x - y\|^2$ on $\mathbb{R}^2 \times \mathbb{R}^2$.

- (a) (**Lower bound**) Show that for any coupling $\pi \in \Pi(\alpha, \beta)$,

$$\int \|x - y\|^2 d\pi(x, y) \geq 1.$$

- (b) (**An optimal coupling**) Define $\pi^\star$ by the disintegration: for $X = (0, s) \sim \alpha$,

$$\pi^\star(\cdot \mid X = (0, s)) = \frac{1}{2} \delta_{(-1, s)} + \frac{1}{2} \delta_{(1, s)}.$$

Check that $\pi^\star \in \Pi(\alpha, \beta)$ and compute its cost. Conclude that the Kantorovich value equals 1.

- (c) (**No Monge minimizer**) Let $T : S_0 \rightarrow S_- \cup S_+$ be a measurable map with $T_\# \alpha = \beta$. Show that

$$\int \|x - T(x)\|^2 d\alpha(x) > 1,$$

hence no Monge map attains the Kantorovich minimum.

- (d) (**Structure of the optimizer and comparison with Brenier**) Describe $\text{supp}(\pi^\star)$ and explain why it is not contained in the graph of any map $x \mapsto T(x)$. Comment on why this does not contradict Brenier's theorem.